

EN.553.724 Probabilistic Machine Learning

Jonathan Y. Ma

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Chapter 1

Introduction

1.1 Probabilistic Modeling and Learning

1.1.1 Classical vs. Modern Methods

Classical probabilistic models (e.g. graphical models) and modern deep generative models differ primarily in how structure is specified.

- **Classical approaches**

- Leverage domain knowledge to specify model structure.
- Work well when the model is correctly specified.
- Provide interpretable probabilistic quantities and uncertainty quantification.

- **Modern approaches**

- Learn highly complex distributions directly from data.
- Require large datasets.
- Often lack interpretability.

- **Goal**

combine structured probabilistic modeling with flexible function approximation.

1.1.2 Learning a Function

Model:

$$y = f(x) + \varepsilon, \quad \mathbb{E}[\varepsilon|x] = 0.$$

Given data $\{(x_i, y_i)\}_{i=1}^N$, choose f_θ .

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N L(f_\theta(x_i), y_i).$$

1.1.3 Maximum Likelihood Interpretation

Assume:

$$\varepsilon \sim \mathcal{N}(0, I).$$

Then:

$$p(y_i|x_i, \theta) \propto \exp\left(-\frac{1}{2}\|y_i - f_\theta(x_i)\|^2\right).$$

Proposition 1.1.1. *Minimizing squared loss is equivalent to maximum likelihood estimation.*

$$\arg \max_{\theta} p(y_{1:N}|x_{1:N}) = \arg \min_{\theta} \sum_{i=1}^N \|y_i - f_\theta(x_i)\|^2.$$

1.1.4 Learning a Distribution

Given $x_1, \dots, x_N \stackrel{\text{iid}}{\sim} \mathbb{P}$:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^N \log p_\theta(x_i) = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N -\log p_\theta(x_i).$$

1.2 Measure-Theoretic Foundations

1.2.1 Discrete and Continuous Probability

- Discrete:

$$\mathbb{P}(A) = \sum_{x \in A} p(x)$$

- Continuous:

$$\mathbb{P}(A) = \int_A p(x) \, dx$$

Unified via:

$$\mathbb{P}(A) = \int_A p(x) \, \lambda(dx).$$

1.2.2 Probability Spaces

Definition 1.2.1. A measurable space is $(\Omega, \mathcal{F})$.

Definition 1.2.2. A probability measure $\mathbb{P}$ satisfies:

- $\mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- countable additivity

If a density exists:

$$\mathbb{P}(A) = \int_A p(x) \, \lambda(dx).$$

1.2.3 Expectation

$$\mathbb{E}[f] = \int f(x) \mathbb{P}(\mathrm{d}x)$$

If density exists:

$$\mathbb{E}[f] = \int f(x)p(x) \lambda(\mathrm{d}x).$$

1.2.4 Radon–Nikodym Derivative

Definition 1.2.3. $\mathbb{P} \ll Q$ if $Q(A) = 0 \Rightarrow \mathbb{P}(A) = 0$.

Theorem 1.2.1 (Radon–Nikodym). *If $\mathbb{P} \ll Q$, then*

$$\int f \mathrm{d}\mathbb{P} = \int f \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q} \mathrm{d}Q.$$

If densities exist:

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}(x) = \frac{p(x)}{q(x)}.$$

1.2.5 Random Variables

Definition 1.2.4. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$.

Distribution:

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A).$$

1.2.6 Conditional Probability

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

$$p(x|y) = \frac{p(x)p(y|x)}{p(y)}$$

1.2.7 Beta-Binomial Model

$$\theta \sim \text{Beta}(\alpha, \beta), \quad X|\theta \sim \text{Binomial}(m+n, \theta).$$

Posterior:

$$\theta|X \sim \text{Beta}(\alpha+m, \beta+n).$$

1.2.8 Conditional Independence

Definition 1.2.5. $X \perp Y|Z$ if

$$p(x, y|z) = p(x|z)p(y|z).$$

1.3 Divergences and Information

1.3.1 Total Variation

Definition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \frac{1}{2} \int |p - q| \lambda(\mathrm{d}x).$$

Proposition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \max_A |\mathbb{P}(A) - Q(A)|.$$

1.3.2 f -Divergence

Definition 1.3.2.

$$D_f(\mathbb{P}||Q) = \mathbb{E}_Q \left[f \left(\frac{p}{q} \right) \right].$$

$$\text{KL}(\mathbb{P}||Q) = \int p \log \frac{p}{q}.$$

1.3.3 Non-Negativity

Theorem 1.3.1 (Jensen).

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Proposition 1.3.2.

$$D_f(\mathbb{P}||Q) \geq 0.$$

Proof.

$$\mathbb{E}_Q f \left(\frac{p}{q} \right) \geq f \left(\mathbb{E}_Q \frac{p}{q} \right) = f(1) = 0.$$

□

1.3.4 Entropy and KL

Definition 1.3.3.

$$H(\mathbb{P}) = - \int p \log p.$$

$$\text{KL}(\mathbb{P}||Q) = -H(\mathbb{P}) - \mathbb{E}_{\mathbb{P}}[\log q].$$

1.3.5 MLE and KL

Proposition 1.3.3.

$$\mathbb{E}_{\mathbb{P}}[-\log p_{\theta}(x)] = \text{KL}(\mathbb{P} \parallel \mathbb{P}_{\theta}) + H(\mathbb{P}).$$

$$\Rightarrow \text{MLE} = \arg \min_{\theta} \text{KL}(\mathbb{P} \parallel \mathbb{P}_{\theta}).$$

1.3.6 KL Direction

$$\text{KL}(\mathbb{P} \parallel Q) \quad \text{mass-covering}$$

$$\text{KL}(Q \parallel \mathbb{P}) \quad \text{mode-seeking}$$

1.3.7 Invariance

Proposition 1.3.4. *For invertible g :*

$$D_f(\mathbb{P} \parallel Q) = D_f(g_{\#}\mathbb{P} \parallel g_{\#}Q).$$

1.4 Optimal Transport and Integral Probability Metrics

1.4.1 Couplings and Wasserstein Distance

Definition 1.4.1. Let $(\Omega, \mathcal{F})$ be a measurable space with metric d , and let μ, ν be probability measures on Ω .

A *coupling* of μ and ν is a probability measure γ on $\Omega \times \Omega$ such that:

$$\gamma(A \times \Omega) = \mu(A), \quad \gamma(\Omega \times A) = \nu(A).$$

Interpretation: a coupling specifies a joint distribution (X, Y) with marginals $X \sim \mu$, $Y \sim \nu$.

Definition 1.4.2. The Wasserstein- p distance is defined as

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \mathbb{E}_{(X, Y) \sim \gamma} [d(X, Y)^p] \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ, ν .

Remark 1.4.1. W_1 is often called the *earth mover's distance*, since it measures the minimal cost of transporting mass from μ to ν .

1.4.2 Integral Probability Metrics

Definition 1.4.3. Let $\mathcal{F}$ be a class of functions $f : \Omega \rightarrow \mathbb{R}$. The integral probability metric (IPM) is

$$d_{\mathcal{F}}(\nu, \mu) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f]|.$$

Lemma 1.4.1. Let $\mathcal{F} = \{f : \|f\|_{\infty} \leq 1\}$. Then

$$\text{TV}(\mu, \nu) = \frac{1}{2} d_{\mathcal{F}}(\nu, \mu).$$

Proof. For any f with $\|f\|_{\infty} \leq 1$,

$$|\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f]| = \left| \int f(x)(\nu - \mu)(dx) \right| \leq \int |f(x)| |\nu - \mu|(dx) \leq \int |\nu - \mu|(dx).$$

Taking the supremum gives

$$d_{\mathcal{F}}(\nu, \mu) \leq 2 \text{TV}(\mu, \nu).$$

Equality is achieved by choosing

$$f(x) = \text{sign}(p(x) - q(x)),$$

which yields

$$\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f] = \int |p(x) - q(x)| dx.$$

□

1.4.3 Kantorovich–Rubinstein Duality

Theorem 1.4.1 (Kantorovich–Rubinstein). Let $\mathcal{F}$ be the set of 1-Lipschitz functions on $\mathbb{R}^d$. Then

$$W_1(\mu, \nu) = d_{\mathcal{F}}(\nu, \mu).$$

Sketch. For any coupling γ and any 1-Lipschitz f ,

$$|f(x) - f(y)| \leq d(x, y).$$

Thus

$$\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f] = \mathbb{E}_{\gamma}[f(Y) - f(X)] \leq \mathbb{E}_{\gamma}[d(X, Y)].$$

Taking infimum over γ gives

$$d_{\mathcal{F}}(\nu, \mu) \leq W_1(\mu, \nu).$$

The reverse inequality follows by constructing an optimal dual potential. □

1.4.4 Remarks on Metrics vs Divergences

- Total variation and Wasserstein are *metrics*.
- f -divergences (e.g. KL) are not metrics:
 - not symmetric
 - do not satisfy triangle inequality
- IPMs provide a unifying view of distances via function classes.

1.5 Maximum Entropy and Exponential Families

1.5.1 Maximum Entropy Principle

Let P be an unknown distribution on Ω . Suppose we are given moment constraints:

$$\mathbb{E}_P[\phi_i(X)] = \mu_i, \quad i = 1, \dots, d,$$

or equivalently

$$\mathbb{E}_P[\phi(X)] = \mu \in \mathbb{R}^d,$$

where $\phi : \Omega \rightarrow \mathbb{R}^d$.

Remark 1.5.1. These constraints alone do not uniquely determine P .

A natural principle is to choose the *least informative* distribution satisfying the constraints.

Definition 1.5.1. The maximum entropy distribution (with respect to a reference measure ν) is

$$\arg \max_P H_\nu(P) \quad \text{s.t.} \quad \mathbb{E}_P[\phi] = \mu,$$

where

$$H_\nu(P) = - \int p(x) \log p(x) \nu(\mathrm{d}x).$$

1.5.2 Derivation via Lagrange Multipliers

Assume Ω is finite for simplicity, and write $p(x) = P(X = x)$.

We maximize

$$- \sum_{x \in \Omega} p(x) \log p(x)$$

subject to:

$$\sum_x p(x) = 1, \quad \sum_x p(x) \phi_i(x) = \mu_i.$$

Define the Lagrangian:

$$\mathcal{L}(p, \lambda_0, \lambda) = - \sum_x p(x) \log p(x) + \lambda_0 \left(\sum_x p(x) - 1 \right) + \sum_{i=1}^d \lambda_i \left(\sum_x p(x) \phi_i(x) - \mu_i \right).$$

Taking derivatives:

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -\log p(x) - 1 + \lambda_0 + \sum_{i=1}^d \lambda_i \phi_i(x).$$

Setting to zero:

$$\log p(x) = \lambda_0 - 1 + \sum_{i=1}^d \lambda_i \phi_i(x).$$

Thus

$$p(x) \propto \exp \left(\sum_{i=1}^d \lambda_i \phi_i(x) \right) = \exp(\langle \theta, \phi(x) \rangle),$$

where $\theta = (\lambda_1, \dots, \lambda_d)$.

This gives the exponential family form.

1.5.3 Exponential Family

Definition 1.5.2. Let $(\mathcal{X}, \nu)$ be a measure space and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$.

The exponential family is

$$p_\theta(x) = \frac{\exp(\langle \theta, \phi(x) \rangle)}{Z(\theta)} = \exp(\langle \theta, \phi(x) \rangle - A(\theta)),$$

where

$$Z(\theta) = \int \exp(\langle \theta, \phi(x) \rangle) \nu(\mathrm{d}x), \quad A(\theta) = \log Z(\theta).$$

Remark 1.5.2. • θ is the natural parameter

- $\phi(x)$ are sufficient statistics
- $A(\theta)$ is the log-partition function

1.5.4 Example: Gaussian from Exponential Family

Let $\phi(x) = (x, x^2)$ on $\mathbb{R}$.

Then

$$p_\theta(x) \propto \exp(\theta_1 x + \theta_2 x^2).$$

Completing the square:

$$\theta_1 x + \theta_2 x^2 = \theta_2 \left(x + \frac{\theta_1}{2\theta_2} \right)^2 - \frac{\theta_1^2}{4\theta_2}.$$

For $\theta_2 < 0$, this is a Gaussian:

$$p_\theta(x) = \mathcal{N}(\mu, \sigma^2), \quad \mu = -\frac{\theta_1}{2\theta_2}, \quad \sigma^2 = -\frac{1}{2\theta_2}.$$

1.5.5 Maximum Entropy Characterization

Theorem 1.5.1. *If there exists θ such that*

$$\mathbb{E}_{p_\theta}[\phi(X)] = \mu,$$

then p_θ is the maximum entropy distribution among all distributions satisfying the moment constraints.

Proof. Let P satisfy $\mathbb{E}_P[\phi] = \mu$. Then

$$\text{KL}(P||p_\theta) = \mathbb{E}_P[\log p(x) - \log p_\theta(x)] \geq 0.$$

Rearranging:

$$-H(P) \geq \mathbb{E}_P[\log p_\theta(x)].$$

Using exponential family form:

$$\mathbb{E}_P[\log p_\theta(x)] = \mathbb{E}_P[\langle \theta, \phi(x) \rangle - A(\theta)] = \langle \theta, \mu \rangle - A(\theta).$$

Thus

$$H(P) \leq A(\theta) - \langle \theta, \mu \rangle = H(p_\theta),$$

so p_θ maximizes entropy. □

1.5.6 Mean Parameter Mapping

Define the mapping

$$\nabla A(\theta) = \mathbb{E}_{p_\theta}[\phi(X)].$$

Definition 1.5.3. Let

$$\mathcal{M} = \{\mu \in \mathbb{R}^d : \exists P \text{ such that } \mathbb{E}_P[\phi(X)] = \mu\}$$

be the set of achievable mean parameters.

Definition 1.5.4. The statistics $\{\phi_i\}_{i=1}^d$ are *minimal* if

$$\{1, \phi_1, \dots, \phi_d\} \text{ are linearly independent.}$$

Proposition 1.5.1. *If ϕ is minimal, then:*

- $\nabla A : \Theta \rightarrow \mathcal{M}$ is injective
- $\nabla A : \Theta \rightarrow \mathcal{M}^\circ$ is surjective (onto the interior)

Sketch. If ϕ is minimal, then $A(\theta)$ is strictly convex:

$$\nabla^2 A(\theta) = \text{Cov}_{p_\theta}(\phi(X)) \succ 0.$$

Strict convexity implies injectivity of ∇A .

Surjectivity onto the interior follows from convex duality and properties of cumulant generating functions. □

1.5.7 Convexity and Geometry

Proposition 1.5.2. *The log-partition function satisfies:*

$$\nabla^2 A(\theta) = \text{Cov}_{p_\theta}(\phi(X)) \succeq 0.$$

Remark 1.5.3. • $A(\theta)$ is convex

- strictly convex if ϕ is minimal
- exponential families form a convex dual pair:

$$\theta \leftrightarrow \mu = \nabla A(\theta)$$

1.5.8 Non-Minimal Case

If ϕ is not minimal, then there exists $v \neq 0$ such that

$$\langle v, \phi(x) \rangle = c \quad \forall x.$$

Then

$$p_{\theta+tv}(x) = p_\theta(x),$$

so the parametrization is not identifiable, and ∇A is not injective.